Practice workbook · Describe a trend or compare data sets and conclude

Describing the pattern in one or more data series from a graph, table or chart, comparing groups or conditions, and drawing a conclusion the data support.

27 question parts: 1 worked-example retry and 26 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-DESCRIBE-COMPARE-TRENDS

1. Question 4 · part a.iv · 2 marks

June 2024 Paper 1R

Worked-example retry · Attempt part a.iv only.

4 (a) The table gives information on how the time taken to digest starch by an enzyme varies at different pH levels.

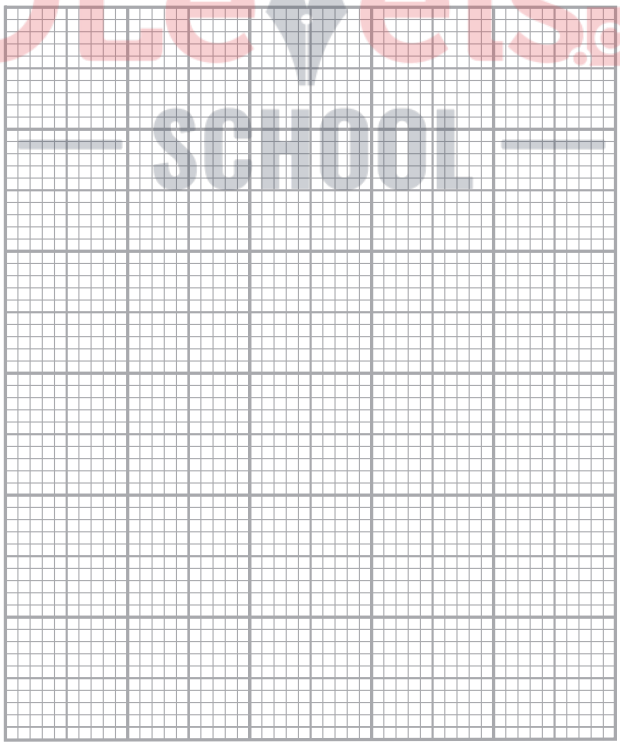
pH	Time taken to digest starch in minutes
2	60
4	45
6	20
8	35
10	55

(i) Plot a graph using the data in the table.

(3)

(ii) Draw a curve of best fit.

(1)



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1. Question 4 · part a.iv · 2 marks

June 2024 Paper 1R

Worked-example retry · Attempt part a.iv only.

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(iii) Use the graph to determine the optimum pH for the digestion of starch.

(1)

(iv) Describe the trend shown by the graph.

(2)

(v) Explain why the time taken to digest starch is different at pH 2 compared with the time taken at pH 6.

(3)

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(vi) Name a part of the body where starch digestion takes place.

(1)

(b) Describe a test for starch in a sample of food.

(2)

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(Total for Question 4 = 13 marks)

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1. Question 4 · part a.iv · 2 marks · Working and answer

Working and answer

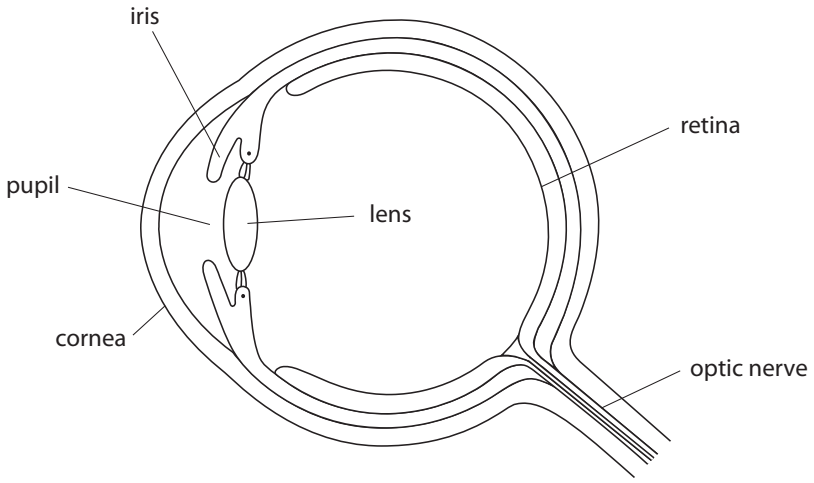
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2. Question 3 · part b.iv · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.iv only.

3 The human eye contains structures that help to focus light on the retina.



(a) Which structures help to focus light on the retina?

(1)

- ☒ **A** cornea and lens
- ☒ **B** iris and cornea
- ☒ **C** iris and pupil
- ☒ **D** pupil and lens

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2. Question 3 · part b.iv · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.iv only.

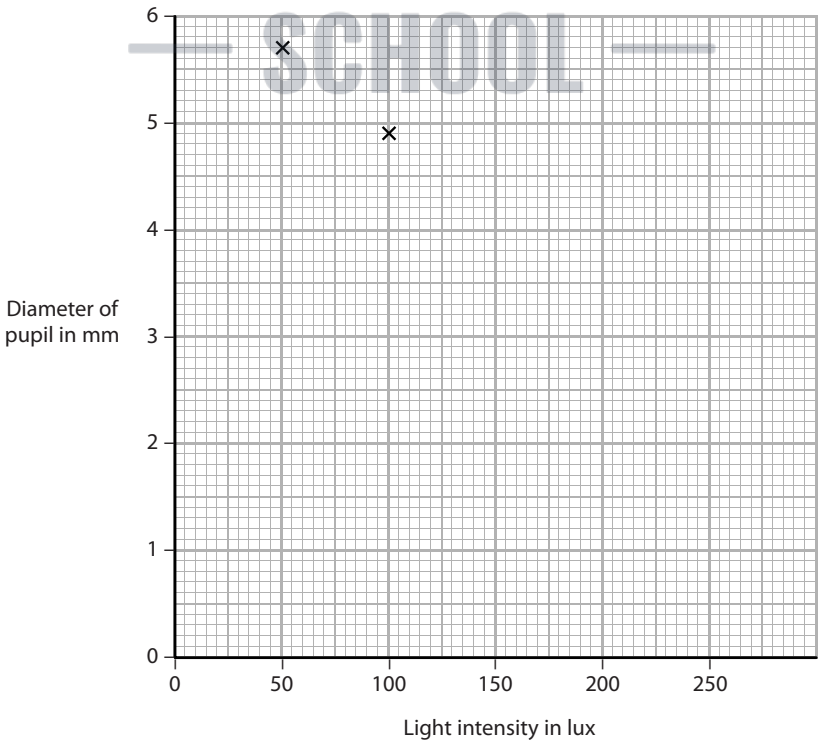
(b) The intensity of light entering the eye can be measured in a unit called lux.

The table shows the results of an investigation on how light intensity affects the diameter of the pupil in the eye.

Light intensity in lux	Diameter of pupil in mm
50	5.7
100	4.9
150	2.7
200	3.2
250	2.3

(i) Complete the graph by plotting the results for 150, 200 and 250 lux. (1)

(ii) Draw a straight line of best fit. (1)



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2. Question 3 · part b.iv · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.iv only.

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(iii) Explain which result is anomalous.

(2)

(iv) State the effect of light intensity on the diameter of the pupil.

(1)

(c) Short-sightedness is a common eye condition.

Explain how short-sightedness affects vision.

(2)

(Total for Question 3 = 8 marks)

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Working and answer

3. Question 6 · part a.i · 1 mark

June 2019 Paper 1

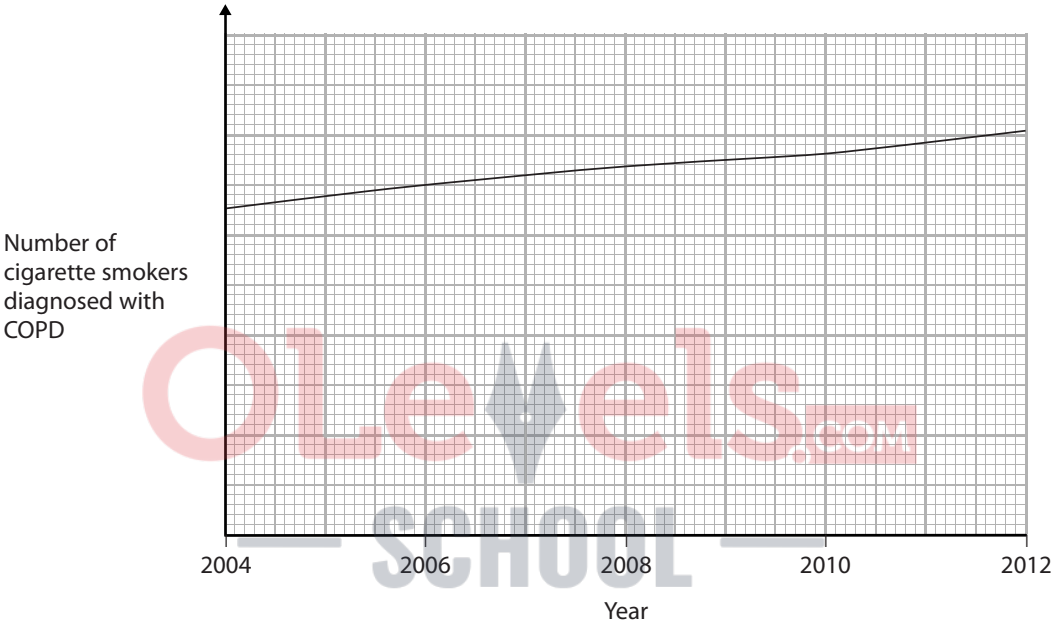
Further practice · Attempt part a.i only.

6 (a) Chronic obstructive pulmonary disease (COPD) is the name given to a group of diseases that affect the breathing system.

These diseases include chronic bronchitis and emphysema.

Cigarette smoking is the main cause of COPD.

The graph shows the number of cigarette smokers diagnosed with COPD in the UK over a period of eight years.



(i) Describe the overall trend in cigarette smokers diagnosed with COPD.

(1)

(ii) Describe what further information is required to help form the conclusion that cigarette smoking is the only cause of COPD.

(2)

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3. Question 6 · part a.i · 1 mark · Working and answer

Working and answer

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4. Question 3 · part a.ii · 2 marks

November 2020 Paper 1

Further practice · Attempt part a.ii only.

3 There are many factors that affect the height of a human.

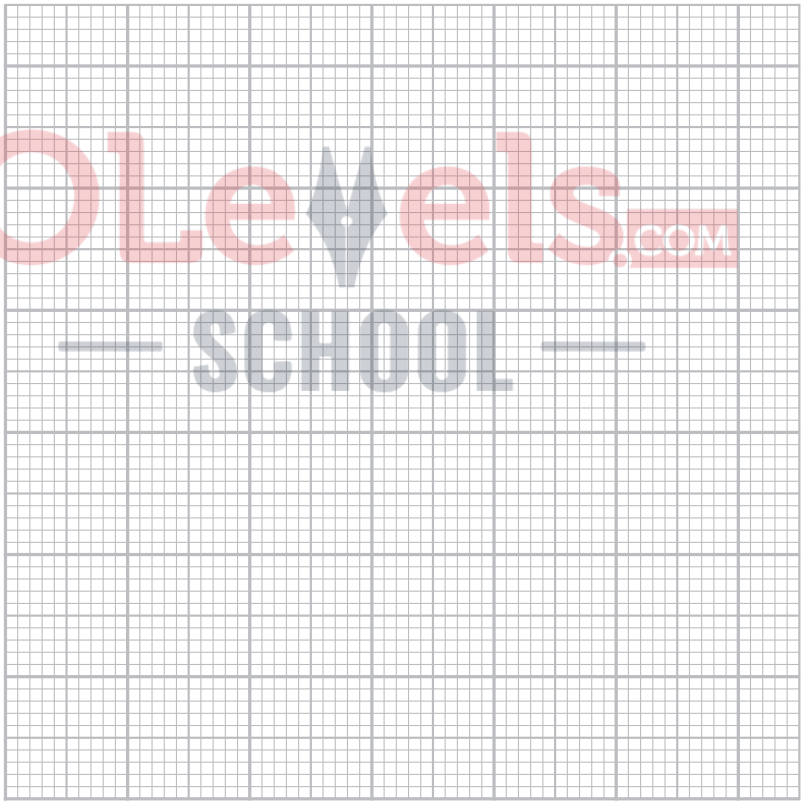
(a) The heights of a group of students are measured.

The table shows the results.

Height in cm	150–151	152–153	154–155	156–157	158–159	> 160
Number of students	4	14	25	20	10	4

(i) Plot a bar chart of the results.

(5)



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4. Question 3 · part a.ii · 2 marks

November 2020 Paper 1

Further practice · Attempt part a.ii only.

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(ii) Describe the trend shown in this bar chart.

(2)

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(b) State three factors that could affect the heights of the students.

(3)

1 .....

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2 .....

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3 .....

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(Total for Question 3 = 10 marks)

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Working and answer

5. Question 3 · part c.i · 2 marks

January 2022 Paper 2

Further practice · Attempt part c.i only.

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3 The skin is important in the control of body temperature.

(a) State two other functions of the skin.

(2)

1 .....

.....

2 .....

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(b) As a result of vigorous exercise, the colour of the skin changes.

Explain why there is this change in skin colour.

(4)

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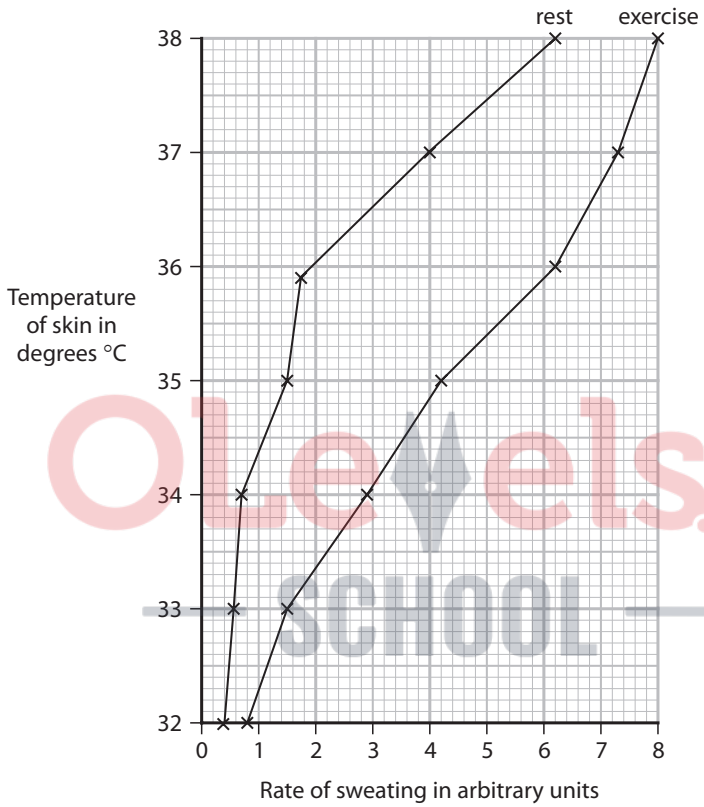
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5. Question 3 · part c.i · 2 marks

January 2022 Paper 2

Further practice · Attempt part c.i only.

(c) The graph shows how the temperature of the skin varies as the rate of sweating changes in a person at rest and during exercise.



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5. Question 3 · part c.i · 2 marks

January 2022 Paper 2

Further practice · Attempt part c.i only.

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(i) Describe how the skin temperature varies as the rate of sweating changes for the person at rest.

(2)

(ii) Suggest a method that could be used to measure the rate of sweating.

(2)

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(d) Explain the importance of maintaining a constant body temperature.

(3)

(Total for Question 3 = 13 marks)

5. Question 3 · part c.i · 2 marks · Working and answer

Working and answer

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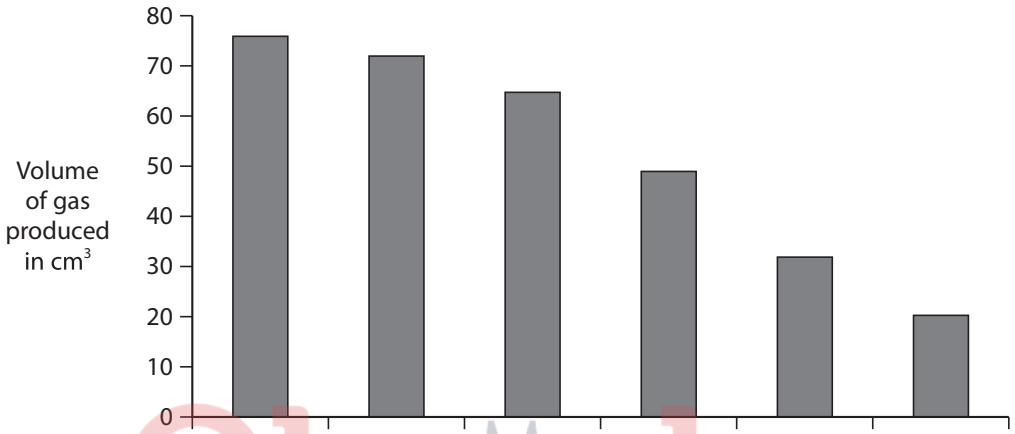
6. Question 3 · part a.iii · 3 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

3 A student investigates a reaction. The reaction is catalysed by an enzyme and produces a gas. The student collects the gas for 10 minutes and measures the volume of the gas. The student repeats this every 10 minutes for one hour.

The bar chart shows the student's results.



(a) (i) Add a suitable labelled scale to the *x*-axis.

(2)

(ii) Estimate the total volume of gas produced after 30 minutes.

(2)

volume of gas = ... cm^3

(iii) Describe the trend shown by the results.

(3)

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6. Question 3 · part a.iii · 3 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

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(iv) Explain the student's results obtained in the investigation.

(3)

(b) Explain the effect on the results if the student had added a competitive inhibitor at the start of the investigation.

(2)

(Total for Question 3 = 12 marks)

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6. Question 3 · part a.iii · 3 marks · Working and answer

Working and answer

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7. Question 4 · part b.i · 2 marks

June 2023 Paper 1

Further practice · Attempt part b.i only.

4 The eye contains receptors that can detect light.

(a) Name two other senses.

(2)

1 .....

2 .....

(b) The table shows how the focal length of the eye lens changes as the thickness of the lens changes.

Thickness of lens in mm	Focal length in mm
10	12.5
9	14.0
8	15.5
7	18.0
6	21.0
5	25.0

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(i) Describe the relationship between the thickness of the lens and its focal length.

(2)

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(ii) Give the effect on the diameter of the lens as its thickness increases.

(1)

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7. Question 4 · part b.i · 2 marks · Working and answer

Working and answer

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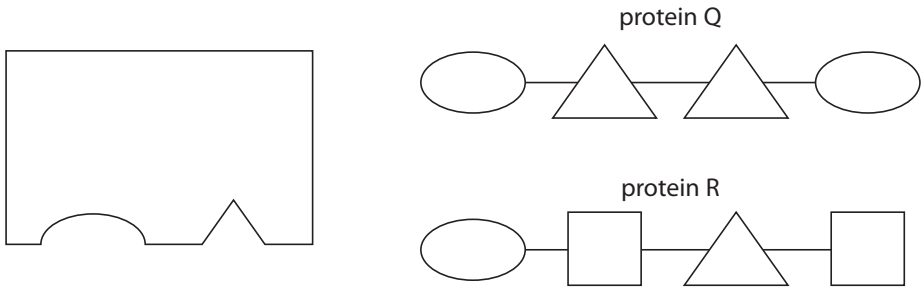
8. Question 2 · part b.i · 3 marks

June 2023 Paper 2R

Further practice · Attempt part b.i only.

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2 (a) The diagram represents an enzyme, known as enzyme P, and two named substrates.



Enzyme P **Substrates**

(i) Use the diagram and words from the box to complete the passage about enzyme P.

(3)

amino acids	amylases	fatty acids	proteases	protein Q	protein R	starch	sugar

Enzyme P belongs to a group of enzymes called ..... .

This enzyme breaks down ..... in food

to ..... .

(ii) Give one location where enzyme P is found in the digestive system.

(1)

.....

(iii) Name the molecule that codes for the production of enzyme P.

(1)

.....

8. Question 2 · part b.i · 3 marks

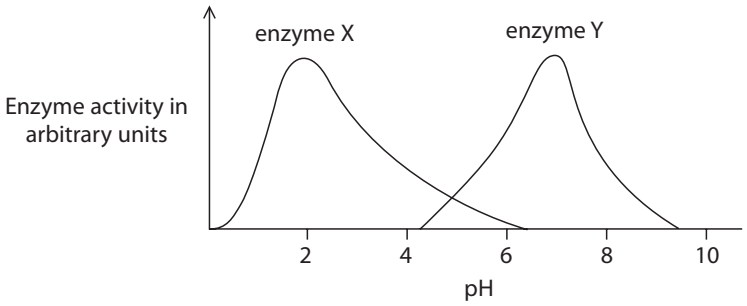
June 2023 Paper 2R

Further practice · Attempt part b.i only.

(b) A student investigates the effect of pH on the activity of two enzymes, X and Y.

The student measures the activity of each enzyme once, using different pH values.

The graph shows the results of their investigation.



(i) Describe how the activity of enzyme X changes with pH.

(3)

(ii) Explain why no activity is shown for enzyme Y at pH2.

(2)

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8. Question 2 · part b.i · 3 marks · Working and answer

Working and answer

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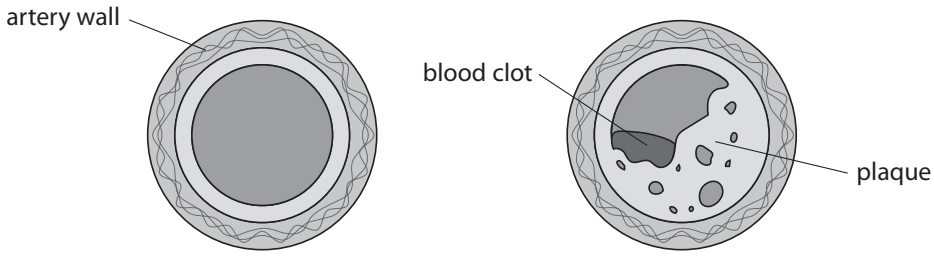
9. Question 6 · part b.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part b.i only.

6 (a) Cholesterol is a fatty substance that can build up inside arteries forming waxy deposits called plaques.

The diagram shows a healthy artery and an artery with a plaque.



healthy artery artery with a plaque

(Source: adapted from https://www.dreamstime.com/artery-plaque-blood-clot-medical-illustration-diseased-compared-to-healthy-cross-section-diagram-shows-image120895536)

Explain how plaques might affect the heart of an individual.

(4)

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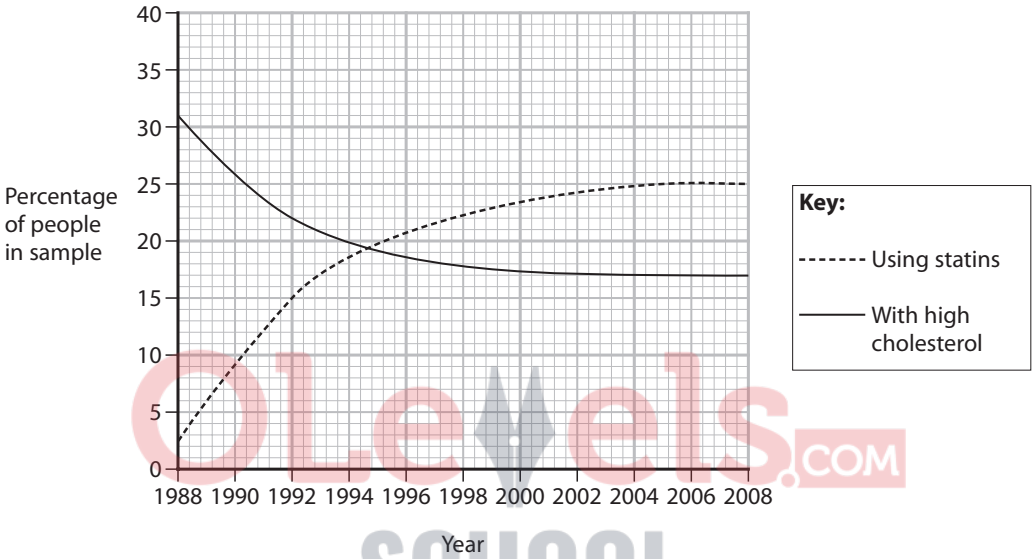
9. Question 6 · part b.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part b.i only.

(b) Scientists investigated the effect of statins on cholesterol levels in a sample of people.

The graph shows the results of their investigation.



(Source: https://www.health.harvard.edu/heart-health/trends-in-high-cholesterol-and-statin-use)

(i) Give the relationship between the use of statins and the percentage of people with high cholesterol levels.

(1)

.....

.....

(ii) Calculate the ratio of the percentage of people with high cholesterol levels in 1988 compared with the percentage of people with high cholesterol levels in 2005.

Give your answer in the form of 1 : n

(3)

ratio = ...

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9. Question 6 · part b.i · 1 mark · Working and answer

Working and answer

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10. Question 1 · part c.i · 2 marks

June 2024 Paper 2

Further practice · Attempt part c.i only.

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1 (a) The table shows some diseases.

Complete the table by writing in the type of pathogen that causes each disease.

(3)

Disease	Type of pathogen
cholera	
AIDS	
athlete's foot	

(b) The passage is about gonorrhoea.

Complete the passage using terms from the box.

(4)

hormonal	antibiotics	contaminated water	virus
sexual intercourse	painkillers	bacterium	barrier

The pathogen that causes gonorrhoea is a ..... .

The pathogen passes from one person to another by ..... .

Gonorrhoea can be prevented by using a ..... method of contraception.

A person with gonorrhoea can be prescribed ..... to kill the pathogen causing the disease.

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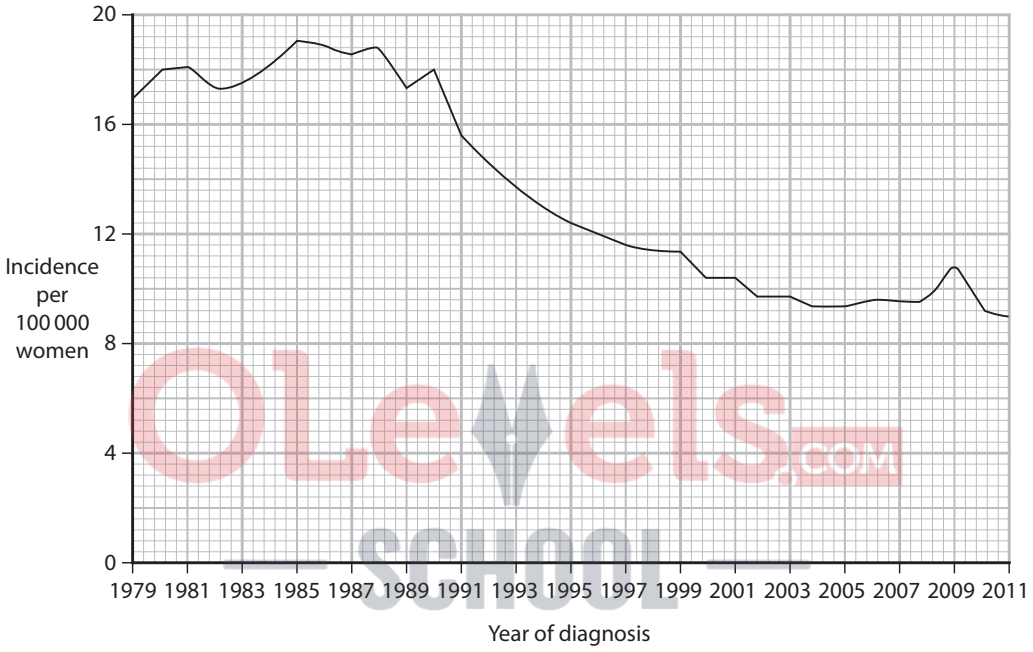
10. Question 1 · part c.i · 2 marks

June 2024 Paper 2

Further practice · Attempt part c.i only.

(c) Cervical cancer is a disease that can be caused by a virus.

The graph shows the incidence of cervical cancer per 100 000 women in the United Kingdom between 1979 and 2011.



(Source: https://www.chattopadhyay.co.uk/cervical-cancer/)

(i) Describe the trends shown by the graph from 1979 to 2011.

(2)

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10. Question 1 · part c.i · 2 marks

June 2024 Paper 2

Further practice · Attempt part c.i only.

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(ii) Calculate the difference in the incidence per 100 000 of women with cervical cancer between 1979 and 2011.

(2)

difference = ... per 100 000

(d) Cervical cancer that is caused by a virus can now be prevented by vaccination.

The table gives information about vaccination.

Which row gives correct information about vaccination?

(1)

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	Contents of vaccination	Response by the body	How the pathogen is destroyed
☒ **A**	antibodies	antigens produced	antigens destroy pathogen
☒ **B**	antibodies	white blood cells produced	white blood cells destroy pathogen
☒ **C**	antigens	antibodies produced	antibodies destroy pathogen
☒ **D**	antigens	platelets produced	platelets destroy pathogen

(Total for Question 1 = 12 marks)

Working and answer

11. Question 2 · part a.iii · 1 mark

November 2024 Paper 1

Further practice · Attempt part a.iii only.

2 A student investigates the activity of an enzyme on fat found in milk.
This is the student's method.

- pour milk and enzyme solution into a test tube
- determine the pH of the solution every minute for 10 minutes

The table shows the student's results.

Time in minutes	pH of solution
1	6.8
2	6.4
3	6.0
4	5.8
5	5.6
6	5.2
7	5.0
8	4.8
9	4.6
10	4.6

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11. Question 2 · part a.iii · 1 mark

November 2024 Paper 1

Further practice · Attempt part a.iii only.

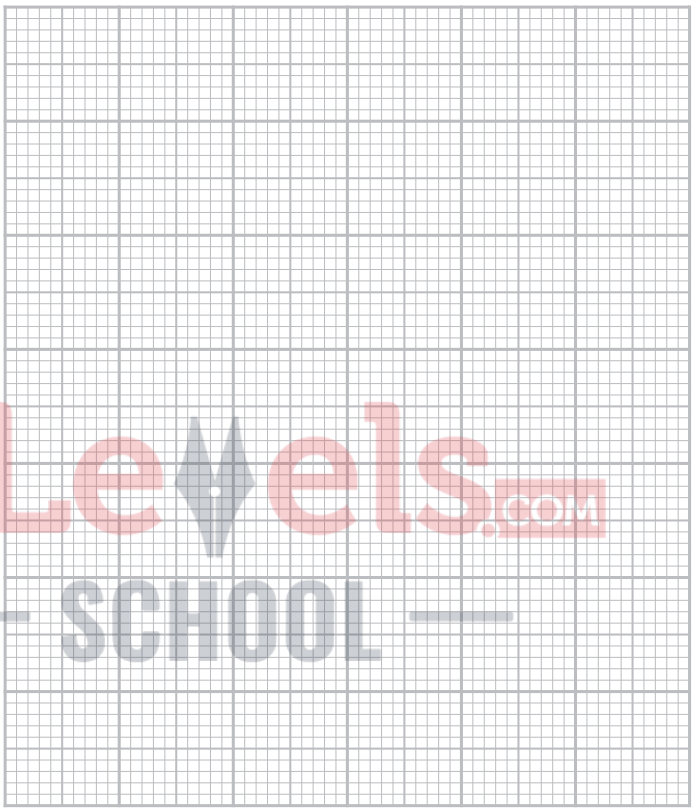
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(a) (i) Plot a graph of the student's results.

(3)



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(ii) Draw a line of best fit.

(1)

(iii) Give the trend shown by the graph.

(1)

.....

.....

11. Question 2 · part a.iii · 1 mark

November 2024 Paper 1

Further practice · Attempt part a.iii only.

(b) Explain the difference in the pH at 1 minute and the pH at 10 minutes.

(3)

(c) Name the independent variable in this investigation.

(1)

(d) Give two variables that the student should control in this investigation.

(2)

1 .....

2 .....

(e) State how the student could improve the reliability of their results.

(1)

(Total for Question 2 = 12 marks)

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11. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

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12. Question 4 · part a.i · 2 marks

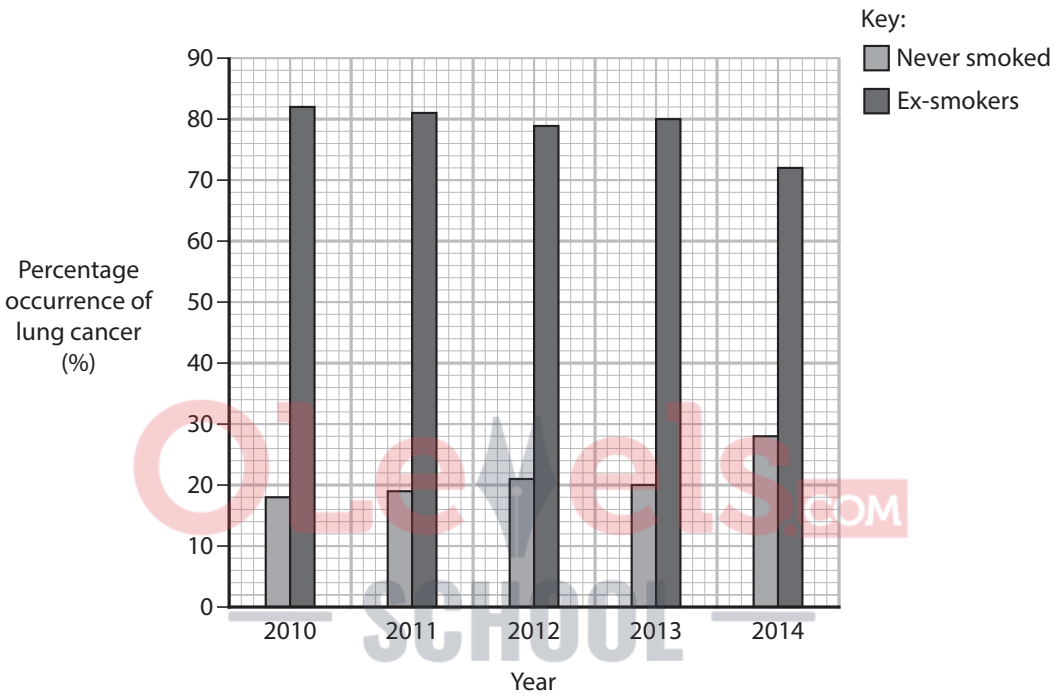
June 2025 Paper 2R

Further practice · Attempt part a.i only.

4 (a) Cigarette smoking is the biggest cause of lung cancer worldwide.

The bar chart shows the percentage occurrence of lung cancer, from 2010 to 2014, in ex-smokers and in people who have never smoked cigarettes.

People who still smoke are not included in the bar chart.



(Source adapted from: https://www.thetimes.co.uk/article/pollution-blamed-for-lung-cancer-in-people-who-have-never-smoked-r52fr7m89)

(i) Describe the trend in the percentage occurrence of lung cancer in ex-smokers, from 2010 to 2014.

(2)

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12. Question 4 · part a.i · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.i only.

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(ii) Calculate the ratio of the percentage occurrence of lung cancer in ex-smokers to people who have never smoked, in 2012.

Give your answer in the form n : 1

(2)

ratio = ... : 1

(iii) Suggest two reasons for the difference in the percentage occurrence of lung cancer, from 2010 to 2014, in people who have never smoked, compared with ex-smokers.

(2)

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(iv) Name the substance in cigarette smoke that causes lung cancer.

(1)

12. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

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13. Question 9 · part b.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part b.iii only.

9 Aspirin is a painkiller, but can also be prescribed to people who are at risk of having a stroke or a heart attack.

Aspirin dissolves blood clots and is also an enzyme inhibitor, which reduces the risk of more blood clots forming in blood vessels.

(a) (i) Describe how a blood clot forms in a blood vessel.

(3)

(ii) Explain how a blood clot in the coronary artery increases the risk of a heart attack.

(3)

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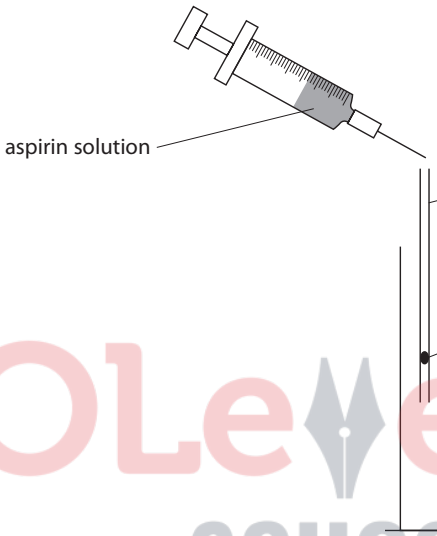
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13. Question 9 · part b.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part b.iii only.

(b) A student uses this apparatus to investigate the effect of aspirin on the rate of blood flow through blood vessels.



The student uses this method.

- dissolve one aspirin tablet in water
- use a syringe to pass the aspirin solution through a straw containing a blood clot
- record the total volume of aspirin solution in the measuring cylinder every five minutes

The student repeats the method with solutions formed from two aspirin tablets and then from three aspirin tablets.

(i) Give a control variable for this investigation.

(1)

(ii) Describe a suitable control test for this investigation.

(1)

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13. Question 9 · part b.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part b.iii only.

The table shows the student's results.

Time in minutes	Total volume of aspirin solution in the measuring cylinder in cm^3		
	1 aspirin tablet	2 aspirin tablets	3 aspirin tablets
5	0	0	0
10	0	0	0
15	0	0	1
20	0	1	2
25	1	1	3
30	1	2	4
35	2	4	6
40	5	7	9
45	10	12	16
50	17	19	22
55	25	27	31
60	35	37	39

(iii) Explain the pattern shown by the results.

(3)

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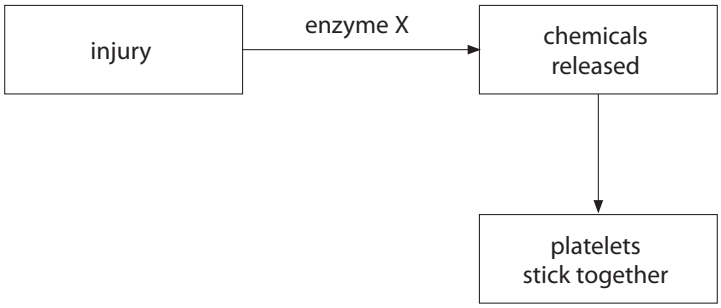
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13. Question 9 · part b.iii · 3 marks

June 2019 Paper 2

Further practice · Attempt part b.iii only.

(c) The diagram shows a pathway involved in blood clotting.



Aspirin is an inhibitor of enzyme X.

Explain how the inhibition of enzyme X reduces the formation of blood clots.

(4)

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(Total for Question 9 = 15 marks)

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13. Question 9 · part b.iii · 3 marks · Working and answer

Working and answer

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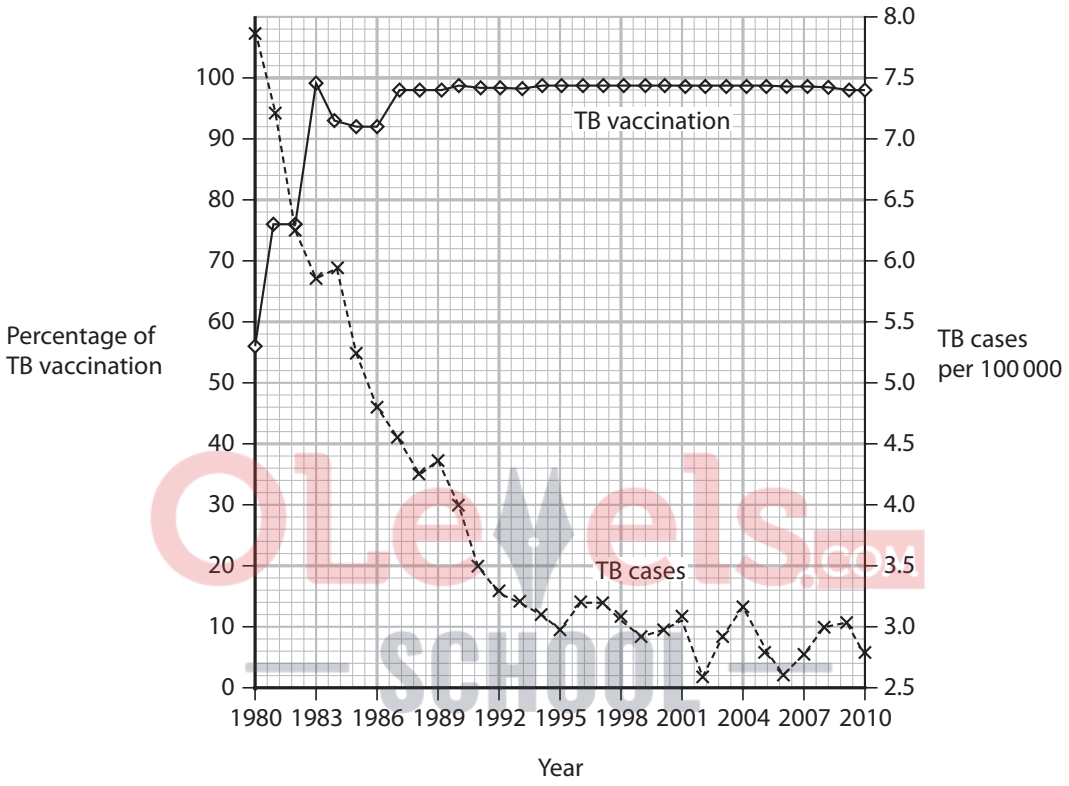
14. Question 4 · part a.ii · 4 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

4 The graph shows the percentage of people vaccinated against tuberculosis (TB) in a country between 1980 and 2010.

It also shows the number of cases of tuberculosis in that country.



(a) (i) Tuberculosis is caused by the same type of organism as typhoid.

What is the type of organism?

(1)

- ☒ **A** bacterium
- ☒ **B** fungus
- ☒ **C** protozoan
- ☒ **D** virus

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14. Question 4 · part a.ii · 4 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

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(ii) Describe the relationship between the percentage vaccination and the number of cases of TB.

(4)

(b) The population of this country in 2010 was 3.45 million.

Calculate the number of cases of tuberculosis in this country in 2010.

(3)

number of cases = ...

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14. Question 4 · part a.ii · 4 marks · Working and answer

Working and answer

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15. Question 3 · part c.i · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.i only.

3 (a) In England, 3 319 266 people have diabetes.

(i) Give this value to 3 significant figures.

(1)

people with diabetes = ...

(ii) In the rest of the United Kingdom, 600 239 people have diabetes.

Give this value in standard form.

(2)

people with diabetes = ...

(iii) Diabetes can be caused by the inability of an organ to produce insulin.

Name the organ that produces insulin.

(1)

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15. Question 3 · part c.i · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.i only.

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(b) The body mass index (BMI) is a measure that uses height and body mass to work out if the body mass is healthy.

The table shows the BMI values and the categories into which a person can be placed.

BMI	BMI category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25.0 to 29.9	overweight
30.0 to 39.9	obese
40.0 and over	morbidly obese

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(i) Calculate the BMI of a person with a body mass of 120 kg and a height of 1.95 m.

Use the equation

$$\text{BMI} = \frac{\text{body mass}}{\text{height}^2}$$

(2)

BMI = ...

(ii) Determine the BMI category that this person belongs to.

(1)

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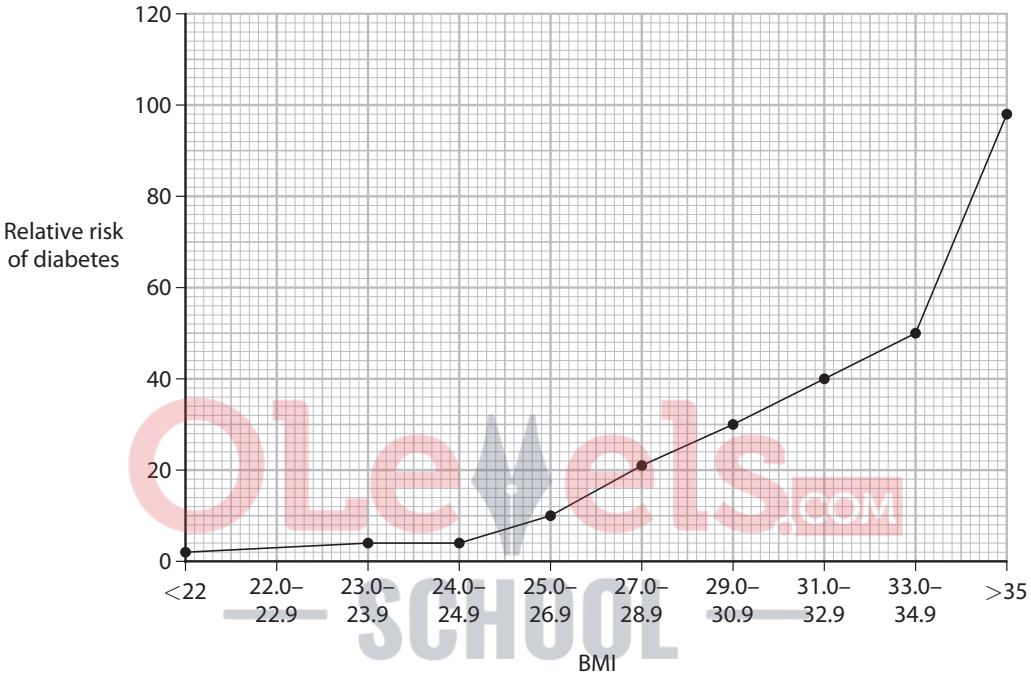
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15. Question 3 · part c.i · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.i only.

(c) The graph shows the relative risk of developing a form of diabetes called type 2 for different BMI values.



(Source: adapted from Colditz GA, et al., *Ann Intern Med.*, 1995; 122:481–486)

(i) Describe the trend shown by the graph.

(2)

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15. Question 3 · part c.i · 2 marks

June 2022 Paper 1R

Further practice · Attempt part c.i only.

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(ii) Give some dietary advice that could help an obese person lower their BMI.

(2)

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(Total for Question 3 = 11 marks)

Working and answer

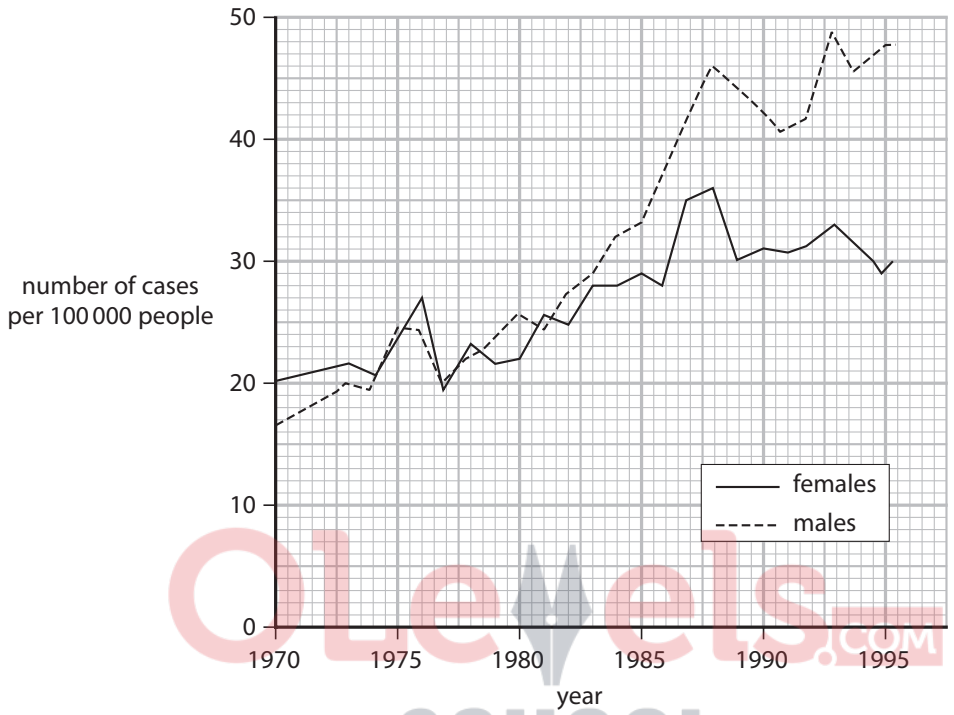
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16. Question 6 · part a.i · 3 marks

November 2021 Paper 1

Further practice · Attempt part a.i only.

6 A form of skin cancer called melanoma has been increasing in both males and females.



(a) (i) Describe the trends shown by the graph.

(3)

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16. Question 6 · part a.i · 3 marks

November 2021 Paper 1

Further practice · Attempt part a.i only.

(ii) Suggest why cancer can occur in the skin.

(2)

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(b) (i) Calculate the mean increase in the number of melanoma cases per 100 000 males per year between 1970 and 1995.

(3)

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mean increase per 100 000 males per year = ...

(ii) In 1970, the population of this area of the world was 7 million.

Calculate the number of females who had melanoma.

(3)

number of females with melanoma = ...

(Total for Question 6 = 11 marks)

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16. Question 6 · part a.i · 3 marks · Working and answer

Working and answer

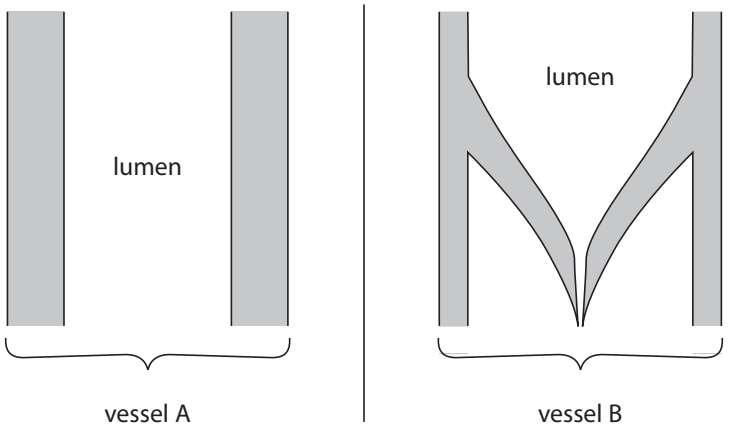
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17. Question 3 · part b.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

3 (a) The diagram shows two types of blood vessel, A and B.



(i) Draw an arrow on the diagram to show the direction of blood flow in vessel B.

(1)

(ii) Identify each type of blood vessel, giving explanations for why there are differences in their structures.

(4)

Vessel A

Vessel B

(iii) State the name of one other type of blood vessel.

(1)

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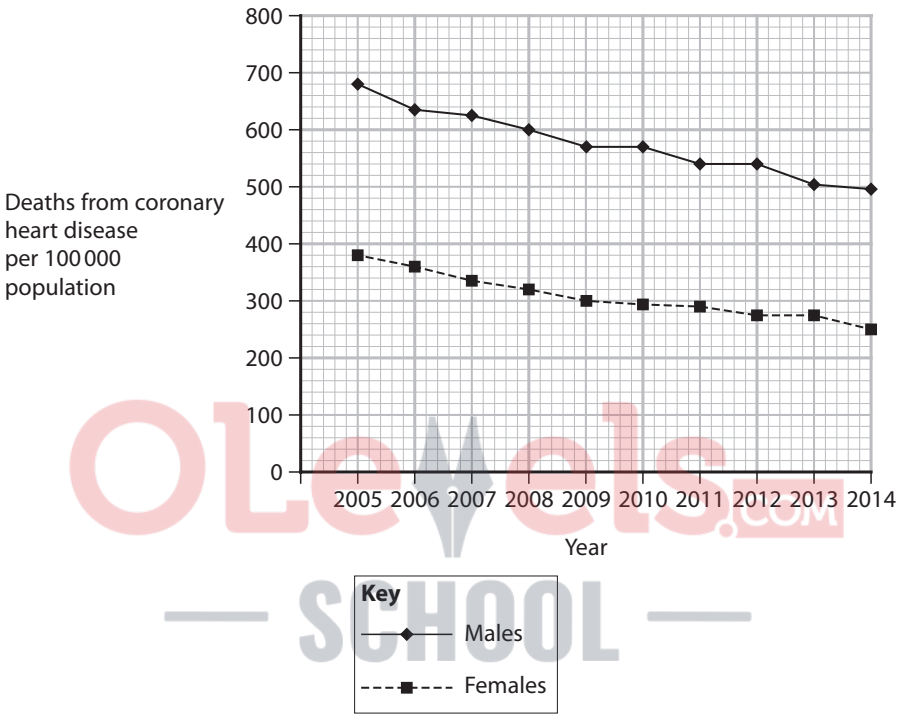
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17. Question 3 · part b.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

(b) Coronary heart disease kills many people each year. The graph shows the number of deaths per 100 000 of the whole population in a part of the world.



(i) Describe the trends shown by the graph.

(3)

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17. Question 3 · part b.i · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

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(ii) The part of the world for which the data was collected had a population of 60 million people in 2006.

Calculate the number of men who died from coronary heart disease in 2006.

(3)

number of men who died = ...

(Total for Question 3 = 12 marks)

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Working and answer

18. Question 4 · part c.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part c.ii only.

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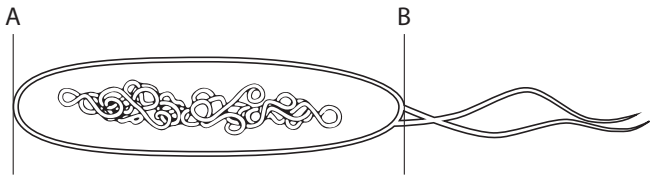
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4 (a) State what is meant by the term **pathogen**.

(1)

.....

(b) The diagram shows a pathogenic bacterium.



(i) Label **three** structures shown on the diagram.

(3)

(ii) The actual length of the bacterium between points A and B is 2.0 µm.

Calculate the magnification of the diagram.

[1 mm = 1000 µm]

(3)

magnification = × .....

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18. Question 4 · part c.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part c.ii only.

(c) The effectiveness of a vaccine in preventing disease was studied over six months.

Two different groups of people were used. One group of people were aged between 45 and 64 years and the second group were aged over 65 years.

The results are shown in the table.

	Percentage effectiveness against disease (%)	
Time since vaccination in months	**45 to 64 years**	**over 65 years**
1	90	80
2	85	78
3	80	75
4	68	58
5	60	50
6	50	40

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18. Question 4 · part c.ii · 3 marks

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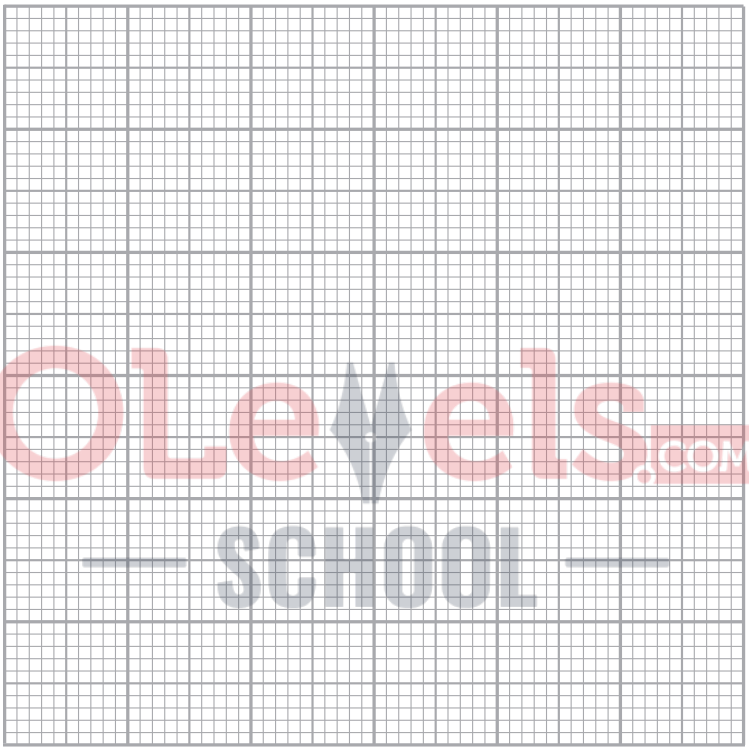
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(i) Plot a graph of the results. Join the points with straight lines.

(5)



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(ii) Describe the trends shown by this graph.

(3)

(Total for Question 4 = 15 marks)

18. Question 4 · part c.ii · 3 marks · Working and answer

Working and answer

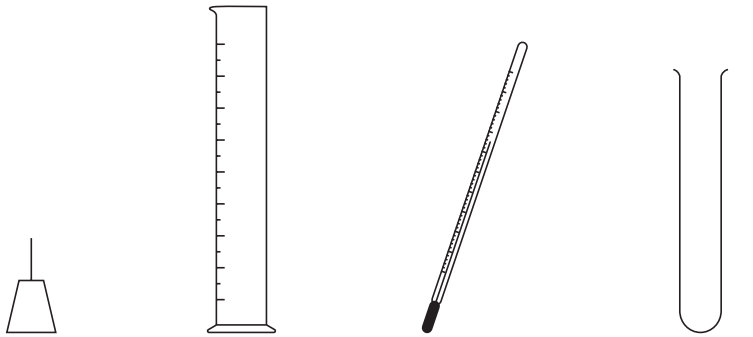
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19. Question 3 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

3 (a) The diagram shows some equipment that can be used to investigate the energy content of different foods.



mounted needle | measuring cylinder | thermometer | test tube

(i) Describe how the equipment can be used to investigate the energy content of different foods.

(5)

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19. Question 3 · part b.i · 2 marks

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Further practice · Attempt part b.i only.

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(ii) Name two variables that need to be controlled in the investigation.

(2)

1 .....

.....

2 .....

.....

(iii) Give two safety precautions needed for this investigation.

(2)

1 .....

.....

2 .....

.....

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19. Question 3 · part b.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part b.i only.

(b) Different people have different energy requirements.

The table shows the daily energy requirements in kilocalories (kcal) of males and females of different ages.

Daily Energy Requirement in kcal

Age of adults in years	Males	Females
19–24	2772	2175
25–34	2749	2175
35–44	2629	2103
45–54	2581	2103
55–64	2581	2079
65–74	2342	1912
75 and over	2294	1840

(Source: adapted from https://propranolols.blogspot.com/2020/02/daily-nutritional-requirements-for_9.html)

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(i) Describe the patterns in the energy requirements for males and for females.

(2)

(ii) Which of these gives a reason for the difference between the energy required by 25-year-old males and the energy required by 25-year-old females?

(1)

- ☒ **A** females use more muscle mass than males
- ☒ **B** females eat less than males
- ☒ **C** males eat less than females
- ☒ **D** males use more muscle mass than females

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19. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

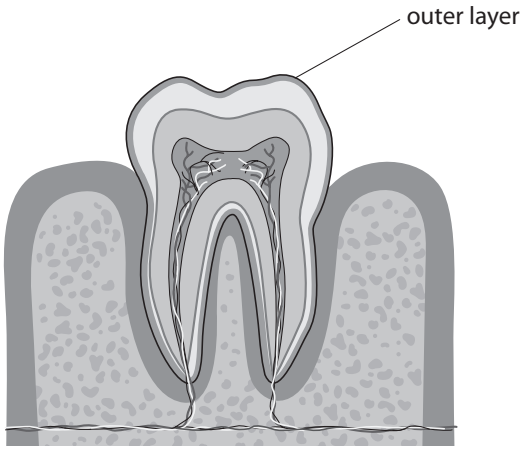
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20. Question 1 · part b.i · 2 marks

June 2019 Paper 2

Further practice · Attempt part b.i only.

1 (a) The diagram shows a section through one type of tooth.



(i) State the name of this type of tooth.

(1)

(ii) Which substance forms the outer layer of the tooth?

(1)

- ☒ **A** bone
- ☒ **B** dentine
- ☒ **C** enamel
- ☒ **D** pulp

(iii) Explain why this type of tooth is more likely to decay than other types of teeth.

(3)

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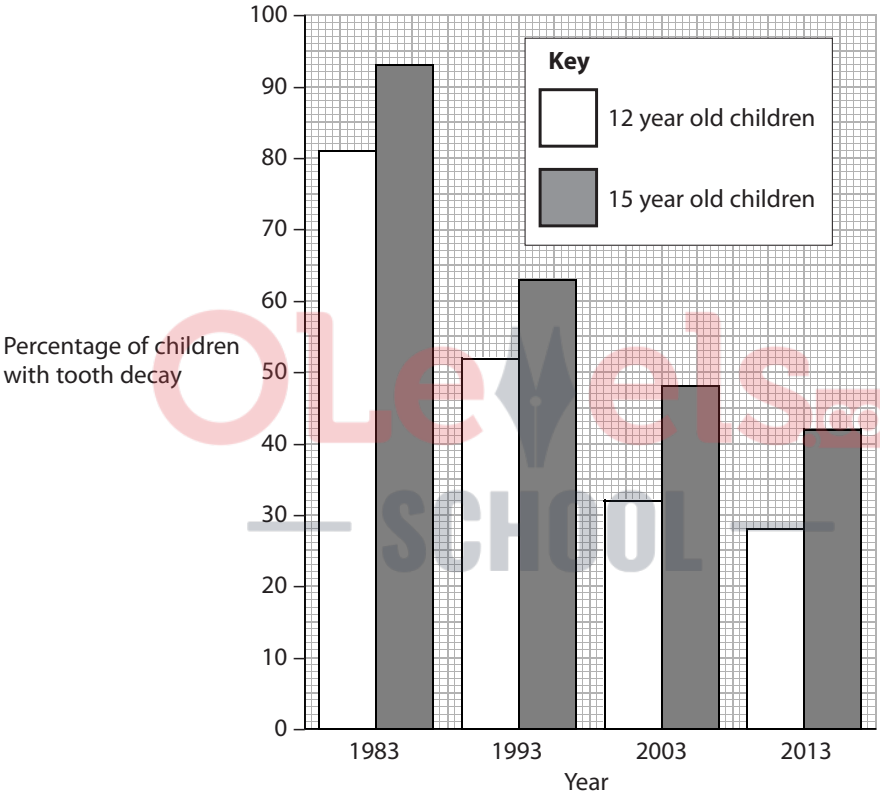
20. Question 1 · part b.i · 2 marks

June 2019 Paper 2

Further practice · Attempt part b.i only.

(b) The bar chart shows the percentage of children with tooth decay.

It gives this information for 12 and 15 year old children over a period of 30 years.



(i) Give two conclusions that can be made from the data in the bar chart.

(2)

1 .....

.....

2 .....

.....

(ii) Suggest a reason for the change in the percentage of children with tooth decay over the 30 year period.

(1)

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(Total for Question 1 = 8 marks)

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20. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

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21. Question 7 · part a.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.ii only.

7 The table gives the percentage of overweight people in different age groups for six countries.

Country	Percentage overweight		Percentage overweight for each age group					
	Males	Females	18–24	25–34	35–44	45–64	65–74	75 years and over
Croatia	67.5	48.3	22.3	38.1	53.7	69.2	73.1	65.7
Malta	66.8	55.2	36.0	52.5	62.7	69.0	74.1	65.3
Poland	64.1	46.7	19.3	38.7	52.8	66.0	73.2	65.5
Portugal	57.6	50.0	23.6	35.1	49.9	63.5	69.3	71.2
Turkey	56.4	56.3	22.4	44.2	64.9	75.4	66.2	56.5
UK	60.2	51.8	29.0	47.3	56.0	64.0	62.8	52.6

(Source from: https://ec.europa.eu/eurostat/statistics-explained/images/5/59/Share_of_overweight_population_by_sex_and_age%2C_2014.png)

(a) (i) The ratio of overweight females to overweight males in the UK is 1:1.2

Calculate the ratio of overweight females to overweight males in Croatia.

Write your answer in the form of 1: n

(2)

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ratio = ...

(ii) Give two trends shown in the data.

(2)

1 .....

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2 .....

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(iii) Give two reasons why the data for a country might be unreliable.

(2)

1 .....

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2 .....

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21. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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22. Question 6 · part a.i · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.i only.

6 *Staphylococcus aureus* (*S. aureus*) and MRSA are microorganisms that cause a wide range of infections in humans.

The table shows the number of deaths from *S. aureus* and from MRSA in England in one year.

The data is for males and females of different age groups.

Age group in years	Number of deaths in one year per million people			
	Males		Females	
	S. aureus	MRSA	*S. aureus*	MRSA
under 45	1.8	0.5	1.4	0.4
45–54	6.0	2.5	4.3	2.4
55–64	15.7	8.2	8.1	4.2
65–74	41.2	26.1	21.9	12.5
75–84	128.6	86.8	81.1	51.6
85 and over	416.6	299.9	200.3	146.4

Source: *Office for National Statistics*

(a) (i) The data shows that there are more deaths caused by *S. aureus* than by MRSA.

Give two other conclusions that can be made from the data.

(2)

1 .....

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2 .....

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22. Question 6 · part a.i · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.i only.

(ii) Calculate the ratio of males aged 85 and over who die from *S. aureus* infection to males under the age of 45 who die from *S. aureus* infection.

(2)

ratio = ...

(iii) The total population in England is 53 million.

Calculate the number of people aged between 65 and 74 who died from *S. aureus*.

(2)

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number = ...

(iv) The number of deaths caused by MRSA is lower than the number of deaths caused by *S. aureus*.

Explain why an infection caused by MRSA is still a major concern.

(2)

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22. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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23. Question 2 · part a.iii · 3 marks

November 2023 Paper 1

Further practice · Attempt part a.iii only.

2 To find out if a person has diabetes, they can be given a glucose tolerance test.

This is how the test is done.

- the person does not eat for 12 hours
- after the 12 hours, blood glucose levels are measured
- the person is then given a solution of glucose to drink
- the blood glucose levels of the person are measured every 30 minutes for 3 hours

The table shows the results for a person without diabetes and a person with diabetes.

Time in minutes	Blood glucose levels in mg per 100 cm^3	
	person without diabetes	person with diabetes
0	65	110
30	120	215
60	130	230
90	95	200
120	80	190
150	65	180
180	65	170

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23. Question 2 · part a.iii · 3 marks

November 2023 Paper 1

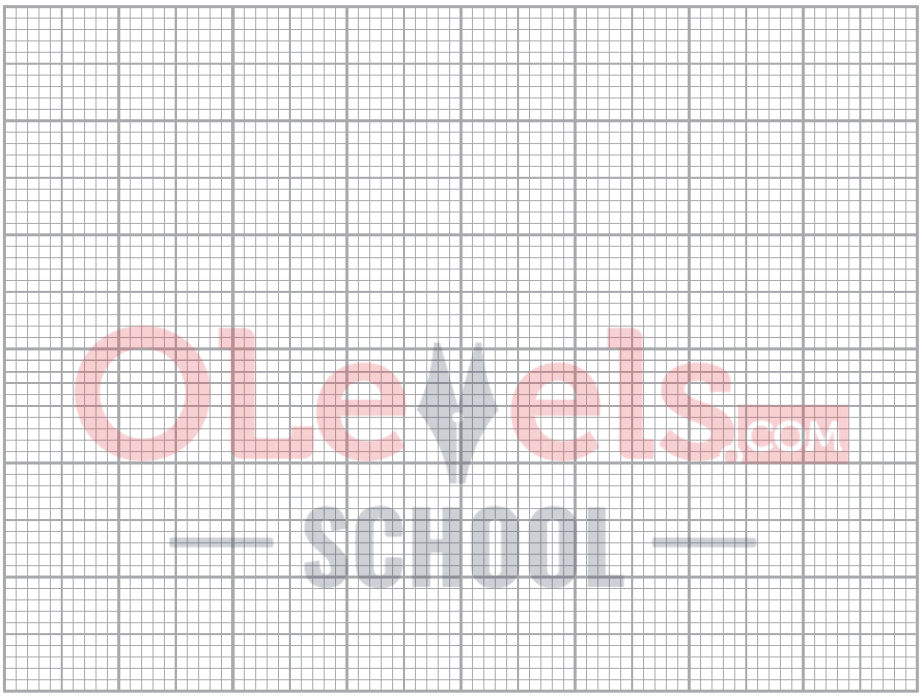
Further practice · Attempt part a.iii only.

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(a) (i) Plot the results on the grid.

Join the points with straight lines.

(6)



(ii) Determine the difference between the highest and lowest blood glucose levels for each person.

(2)

without diabetes = ... mg per 100 cm^3

with diabetes = ... mg per 100 cm^3

23. Question 2 · part a.iii · 3 marks

November 2023 Paper 1

Further practice · Attempt part a.iii only.

(iii) Give **three** differences between the glucose levels for the two people during the three hours.

(3)

1 .....

.....

2 .....

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3 .....

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(b) One symptom of diabetes is glucose in the urine.

Describe how a sample of urine could be tested safely for glucose.

(5)

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(Total for Question 2 = 16 marks)

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23. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

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24. Question 3 · part b.i · 3 marks

November 2023 Paper 1

Further practice · Attempt part b.i only.

3 The table shows the flow of blood to different organs in a person at rest and during exercise.

	Blood flow in cm^3 per minute	
	at rest	during exercise
Heart	250	750
Skeletal muscles	1000	12 500
Skin	400	1900
Gut	1400	600
Brain	750	750
Total flow	5600	17 500

(a) (i) Give the ratio of the blood flow to the heart when at rest and during exercise.

Give your answer in the form 1 : n

(2)

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Ratio = ... : .....

(ii) Calculate the percentage increase in the total flow of blood between rest and exercise.

Give your answer to two significant figures.

(3)

percentage increase = ... %

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24. Question 3 · part b.i · 3 marks

November 2023 Paper 1

Further practice · Attempt part b.i only.

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(iii) Explain why the total blood flow at rest is greater than the flow to the heart, skeletal muscle, skin, gut and brain added together.

(3)

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(b) (i) Describe the main changes between the blood flow values at rest and during exercise.

(3)

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(ii) Explain the change in blood flow rate to the skin when at rest and during exercise.

(4)

(Total for Question 3 = 15 marks)

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24. Question 3 · part b.i · 3 marks · Working and answer

Working and answer

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25. Question 3 · part e · 2 marks

November 2023 Paper 2

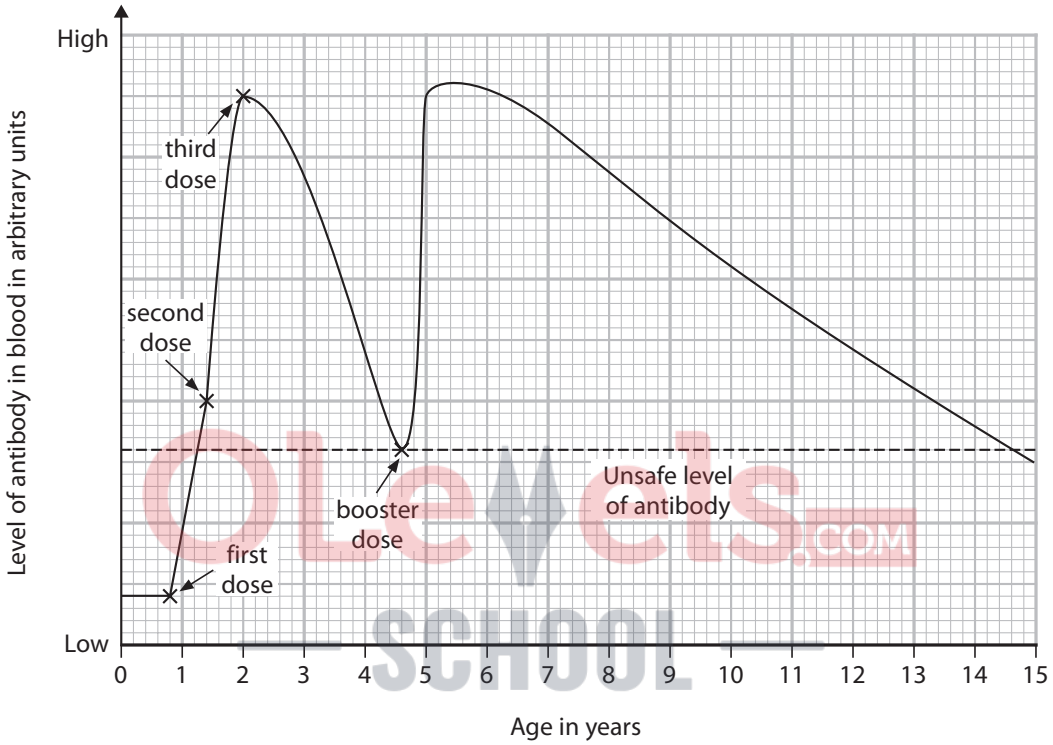
Further practice · Attempt part e only.

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3 A vaccine is used to protect people against a disease.

The graph shows the level of antibodies in the blood of a child given the vaccine.

Three doses of the vaccine were given and then a booster dose was given.



(a) State the general name given to disease causing organisms.

(1)

.....

(b) (i) Determine the time, in years, between the first dose and the third dose of the vaccine.

(1)

time = ... years

(ii) State the type of immunity that vaccination produces.

(2)

.....

.....

P 7 4 4 9 4 A 0 5 2 0

25. Question 3 · part e · 2 marks

November 2023 Paper 2

Further practice · Attempt part e only.

(iii) Name two components that could be in a vaccine that would help to trigger immunity.

(2)

1 .....

2 .....

(c) There are some antibodies in the blood of the child before the first vaccination is given.

Explain why there are antibodies in the blood before vaccination.

(2)

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(d) Explain why a booster dose is necessary.

(5)

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25. Question 3 · part e · 2 marks

November 2023 Paper 2

Further practice · Attempt part e only.

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(e) Give two differences between the effects of the first three doses and the effect of the booster dose.

(2)

1 .....

.....

2 .....

.....

(Total for Question 3 = 15 marks)

Working and answer

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26. Question 3 · part b · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

3 A woman who smokes when she is pregnant risks harming her developing fetus.

The table lists the birth masses of 310 children born to non-smoking mothers or to mothers who smoke.

	Birth masses of children in kg					
	Less than 2.4	**2.4 – 3.1**	**3.2 – 3.5**	**3.6 – 3.9**	**More than 3.9**	**Total number of births in sample**
Number of children born to non-smoking mothers	4	42	70	60	34	210
Number of children born to mothers who smoke	3	24	44	23	6	100

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(a) (i) What is the most common birth mass in kg? **(1)**

- ☒ **A** 2.4 – 3.1
- ☒ **B** 3.2 – 3.5
- ☒ **C** 3.6 – 3.9
- ☒ **D** More than 3.9

(ii) Calculate the percentage of children with a birth mass of 3.6 kg or more in the sample of non-smoking mothers. **(2)**

percentage = ...

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26. Question 3 · part b · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

(b) Use data from the table to deduce what can be concluded about the effect of smoking on birth mass.

(2)

(c) Suggest why smoking affects birth mass.

(5)

(Total for Question 3 = 10 marks)

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26. Question 3 · part b · 2 marks · Working and answer

Working and answer

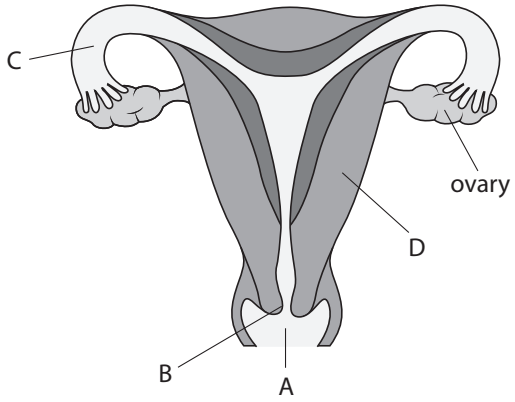
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27. Question 2 · part d.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part d.ii only.

2 (a) The diagram shows some structures of the female reproductive system.



(i) Where are sperm deposited during sexual intercourse?

(1)

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

(ii) Which part of the female reproductive system contracts during birth to push the fetus downwards?

(1)

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

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27. Question 2 · part d.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part d.ii only.

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(iii) The ovaries produce female sex hormones.

The box lists words associated with the female reproductive system.

FSH	wall	fertilisation	oestrogen	ovulation
lining	ADH	follicles	menstruation	

Use words from the box to complete the passage about the ovaries.

(3)

The ovaries produce a hormone called ..... .

This hormone repairs the uterus ..... following menstruation. Once a month, one egg is released from the ovaries. This is known as ..... .

(b) State why a pregnant female should have more calcium in her diet than a female who is not pregnant.

(1)

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(c) Some women breastfeed their babies.

Give one advantage of breastfeeding babies rather than feeding them bottled milk.

(1)

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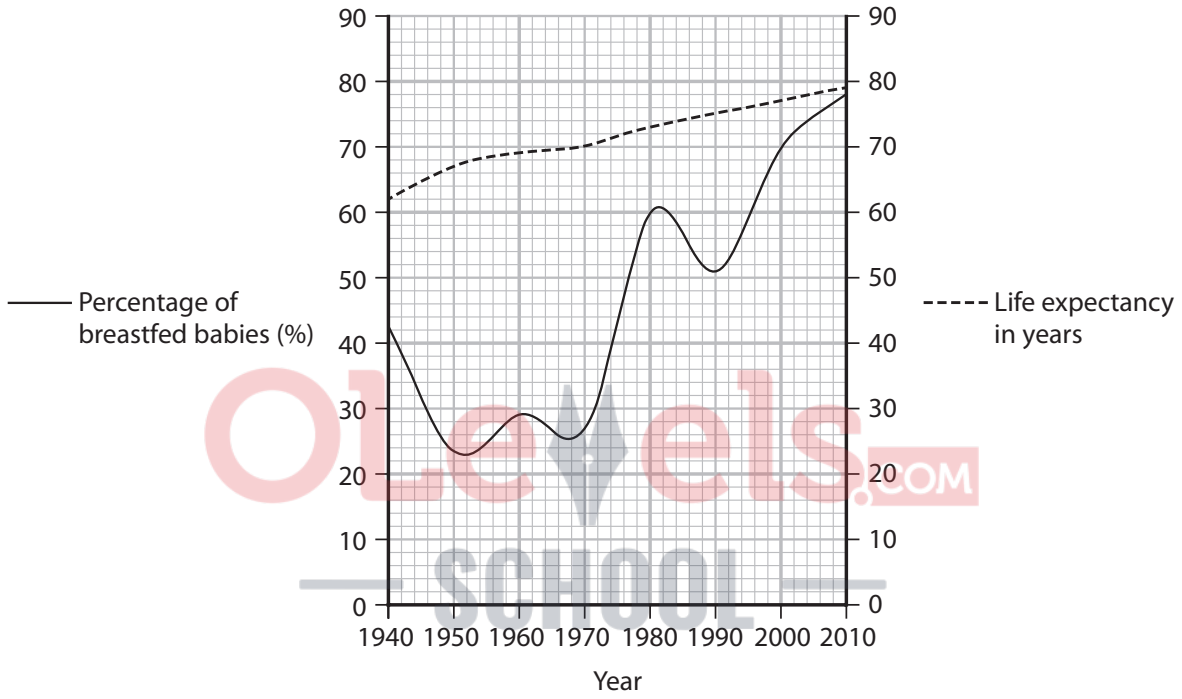
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27. Question 2 · part d.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part d.ii only.

(d) The graph shows the percentage of babies that are breastfed and the life expectancy of breastfed babies from 1940 to 2010 in the UK.



(i) Compare the percentage of babies that were breastfed in 1940 with the percentage of babies that were breastfed in 2010.

(1)

.....

.....

(ii) The graph suggests that breastfeeding has no effect on life expectancy.

Describe how the graph shows this.

(2)

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(Total for Question 2 = 10 marks)

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27. Question 2 · part d.ii · 2 marks · Working and answer

Working and answer

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